

☒ EASY

☐ ALGEBRA

Mixed

10 questions · ~15 min

03

Q1

ALGEBRA

The function f is defined by $fx = 80 - 6x$. What is the value of $f7$?

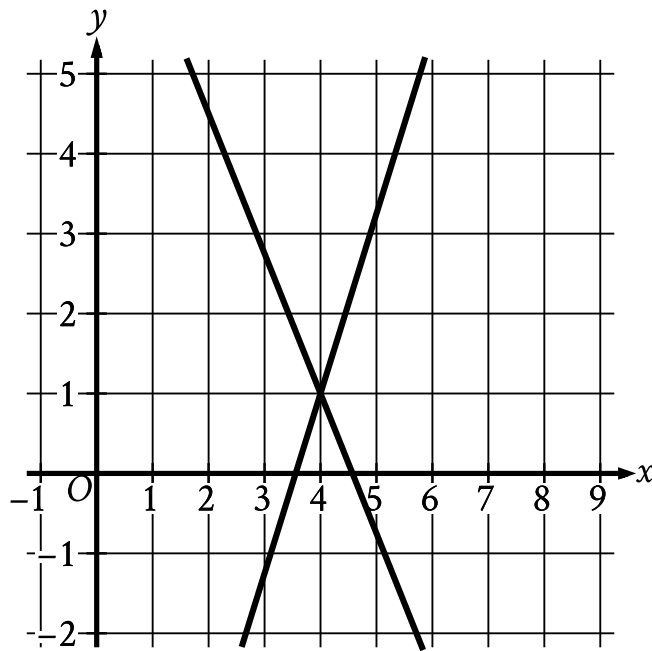
- A. 13
- B. 38
- C. 74
- D. 81

Q2

ALGEBRA

The length, y , of a white whale was 162 centimeters cm when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (4 comma 1)
 - (StartFraction 32 Over 7 EndFraction comma 0)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (StartFraction 32 Over 9 EndFraction comma 0)
 - (4 comma 1)

The graph of a system of linear equations is shown. The solution to the system is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ALGEBRA

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

5 / 8

Marta has 7,500 pesos she will convert to US dollars using a currency exchange service. At this time, the currency exchange rate is 1 peso = 0.075 US dollars. The exchange service will charge Marta a 2% fee on the converted US dollar amount. How many US dollars will Marta receive from the currency exchange after the 2% fee is applied?

- A. \$551.25
- B. \$562.50
- C. \$5,625.00
- D. \$98,000.00

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

The high temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature ($^{\circ}\text{F}$)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F ?

- A. Median of the high temperatures
- B. Mean of the high temperatures
- C. Mode of the high temperatures
- D. Range of the high temperatures

Q8

ADVANCED MATH

Which expression is equivalent to $20w - 4w + 3w$?

- A. $10w$
- B. $13w$
- C. $19w$
- D. $21w$

Q9

ADVANCED MATH

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

- A. 6
- B. 30
- C. 450
- D. 900

Q10

GEOMETRY AND TRIGONOMETRY

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

- A. 9
- B. 70
- C. 315
- D. 560